

Mirza Farhan Bin Tarek

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Ph.D. Candidate in Computer Science — Machine Learning • Statistical Modeling • Health Informatics

Summary

Ph.D. candidate in Computer Science at the University of Delaware, researching machine learning and data-driven modeling in healthcare. Experienced in statistical analysis, large-scale data processing, and applied reinforcement learning. Currently developing methods to reduce hallucination and reward hacking and improve reliability in large language models for clinical decision support.

Education

University of Delaware , Newark, DE Ph.D. in Computer Science; M.Sc. in Computer Science (Completed 2024) <i>Research:</i> Hallucination mitigation in clinical LLMs. <i>Relevant Coursework:</i> Machine Learning, Advanced Deep Learning, Database Systems	2022 – 2027 (<i>Expected</i>)
University of Staffordshire , Stoke-on-Trent, UK Individual Modules, Erasmus+ KA1 Mobility Program <i>Relevant Coursework:</i> Research Methods	2017 – 2018
United International University , Dhaka, Bangladesh B.Sc. in Computer Science and Engineering, <i>Summa Cum Laude</i> <i>Relevant Coursework:</i> Machine Learning, Probability and Statistics, Data Mining	2014 – 2020

Research Experience

Graduate Researcher, University of Delaware • Developed fairness-optimized and reinforcement learning-based models and pipelines for clinical predictions and LLM alignment. Implemented REINFORCE and PPO for policy optimization on clinical decision tasks. • Designed and optimized scalable ML pipelines processing 50K+ patient records on distributed GPU clusters using PyTorch and scikit-learn; reduced training time by 35%. • Led published research on bias mitigation and reward optimization, featured at IEEE CHASE'25 and ACM-BCB'25 . • Contributed as co-author to studies on fairness and clinical prediction accepted at ACM FAccT'23 and the Investigative Ophthalmology & Visual Science Journal . • Served as reviewer for AHLI CHIL'25 and AISTATS'26 .	2022 – 2027
Research Collaborator, Temple University & Wilmington VA Medical Center • Built ML models (Decision Tree, XGBoost, Random Forest) to identify diabetic retinopathy risk in 20K+ patient records. • Building models to identify retinopathy risk from time-series data using the BEHRT algorithm.	2023 –

Selected Projects

TA Dashboard Automation System — FastAPI, Python, MySQL • Designed and deployed a FastAPI-based analytics dashboard for 40+ users; automated TA allocation and generated summary statistics and visual reports from MySQL data.	2023 – 2024
Development of Machine Learning Models and a Product Predictor to Identify Patients at Risk for Diabetic Retinopathy in a Veteran Population — PyTorch, Transformers • Developed a machine learning (ML) model to classify patients into low- and high-risk categories for DR, allowing personalized screening intervals to optimize resource use and focus on high-risk individuals. • Applied statistical modeling and data visualization to analyze model fairness metrics (SPD, AOD) and interpret large-scale datasets. • Analyzed and modeled 20K+ patient records using XGBoost and Random Forest to predict diabetic retinopathy risk; achieved AUROC > 0.8 and deployed interpretable model outputs for clinical decision support.	2023 – 2025

Technical Skills

Programming:	Python, Java, C
Data Analysis:	Pandas, NumPy, SciPy, scikit-learn, Matplotlib, Seaborn
Statistical / ML:	Regression, Hypothesis Testing, Reinforcement Learning
Big Data / Cloud:	PySpark, AWS Sagemaker, AWS Bedrock API
Tools:	Git, Docker, Linux, Windows
Databases & Web:	MySQL, Oracle, REST APIs, Jinja2, HTML/CSS, JavaScript

Selected Publications

- **Tarek, Mirza Farhan Bin**, and Rahmatollah Beheshti. "Reward Hacking Mitigation using Verifiable Composite Rewards." arXiv preprint arXiv:2509.15557 (2025). *Accepted at ACM-BCB'25*.
- **M. F. Bin Tarek**, R. Poulain, and R. Beheshti, "Fairness-Optimized Synthetic EHR Generation for Arbitrary Downstream Predictive Tasks," 2025 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), New York, NY, USA, 2025, pp. 234-244.
- Poulain, R., **Tarek, M. F. B.**, and Beheshti, R. (2023). Improving Fairness in AI Models on Electronic Health Records: The Case for Federated Learning Methods. FAccT '23: Proceedings of the 2023 ACM Conference on Fairness, Accountability, and Transparency. Association for Computing Machinery, 2023, 1599–1608.

Teaching Experience

Instructor & Teaching Assistant, University of Delaware

Aug 2022 – Present

- Instructor for *CISC181 – Intro to Computer Science II* (Summer '23–'25); taught Java and OOP to 30+ students.
- TA for undergraduate courses, including Data Mining and Machine Organization; graded, mentored, and automated grading scripts for assessment analytics.

Lecturer, Brac University, Dhaka, Bangladesh

2019 – 2022

- Lecturer for CSE340: Computer Architecture, CSE461: Introduction to Robotics
- Course coordinator of CSE340: Computer Architecture, where I organized and coordinated exams and updated the course curriculum to meet OBE standards.

Awards

- Awarded NSF Student Travel Award for attending IEEE/ACM Chase'25
- Selected to participate in the Erasmus+ KA1 Mobility Program (2017)
- Completed Coursera Deep Learning Specialization (5-course series), Stanford University (2020)